

SENTIMENT ANALYSIS AND FEATURE EXTRACTION FOR PREDICTING ONLINE

REVIEW RATINGS USING NLP AND SUPERVISED LEARNING

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1. Executive Summary

This project investigates whether textual review content (customer comments) aligns with the star rating assigned by reviewers, and whether we can predict ratings from text using simple, interpretable methods suitable for an MBA-level dissertation. The pipeline uses standard NLP preprocessing, TF-IDF for feature extraction, rule-based sentiment scoring (TextBlob), and a Logistic Regression classifier for supervised prediction. The model achieved ~72% accuracy at predicting star ratings (1–5). The analysis highlights that sentiment polarity correlates strongly with rating — particularly for extreme ratings — and provides business-actionable insights on words and issues driving positive and negative reviews.

2. Background / Problem Statement

Online retailers receive thousands of textual reviews daily. While star ratings provide a quick snapshot, they don't explain *why* customers felt a certain way. This project aims to:

- Measure the sentiment in customer reviews.
 - Extract text features that represent reviews numerically.
 - Build an interpretable model to predict ratings from text.
 - Provide visual and actionable insights for product teams.
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3. Motivation

For business stakeholders, automated understanding of textual feedback saves time and identifies product issues early. This project aims to demonstrate a straightforward, scalable approach that balances interpretability and performance — important in business settings where decisions must be justified to non-technical stakeholders.

4. Data & Preprocessing

Dataset: Uploaded and cleaned notebook: amazon_cleaned.csv (a cleaned version of raw Amazon reviews).

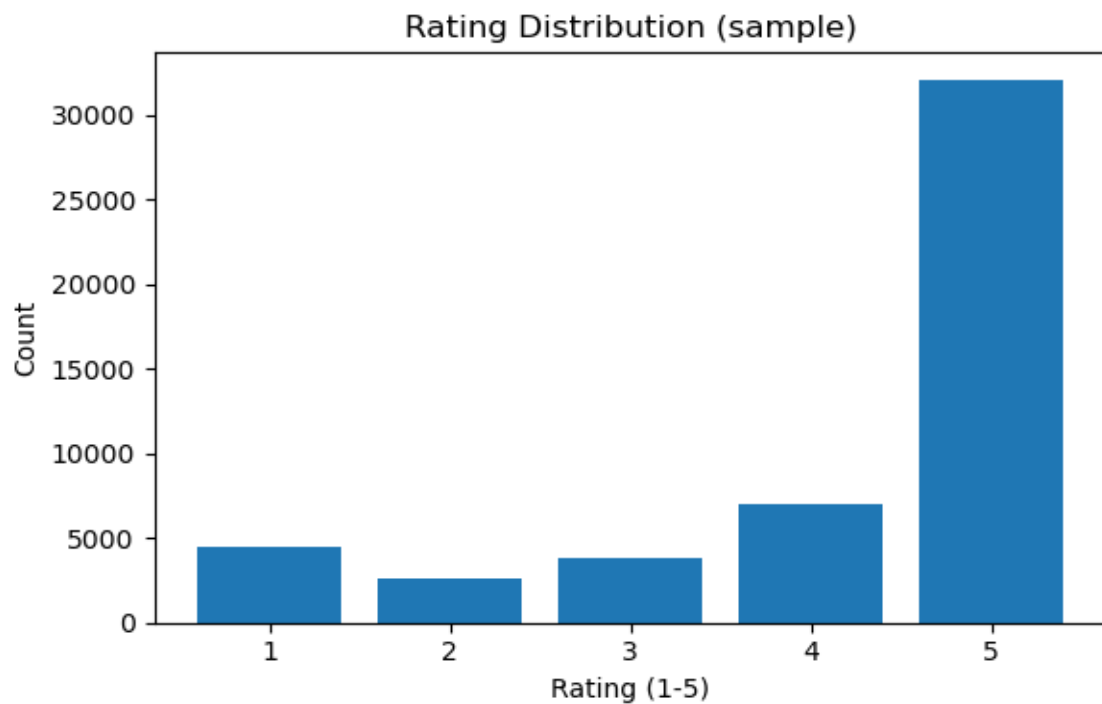
Key columns:

- cleaned_text — cleaned review text (lowercased, punctuation removed, stopwords removed, lemmatized).
- Score — original star rating (1–5).
- txt_len — character length of original review (used for EDA).

Initial observations (from EDA):

- Rating counts are imbalanced (5-star reviews dominate).

Figure 1: Rating Distribution of Amazon Product Reviews



- The distribution clearly shows that 5-star and 4-star reviews form the majority of the dataset, indicating generally high customer satisfaction. Lower ratings (1-star and 2-star) are relatively rare, which can bias model training.
- Average review length ~436 characters; some extreme outliers with very long reviews.

Cleaning approach :

- Convert text to lowercase.
- Remove punctuation, numbers, special characters.
- Tokenize and remove stopwords.
- Lemmatize tokens.
- Recombine tokens into `cleaned_text`.
- Fill missing cleaned texts with empty strings to avoid processing errors.

(All of the above steps are implemented in the notebook cell that defines `clean_text` and applies it to the dataset.)

5. Feature Extraction (TF-IDF)

Method: `TfidfVectorizer(max_features=2000, stop_words='english')`.

Why TF-IDF:

- It balances term frequency (how common a word is in a document) with inverse document frequency (how rare the word is across all documents).
- This reduces weight of common words and highlights distinguishing words that are informative for ratings.

What the code did:

- Created `tfidf` vectorizer and transformed `cleaned_text` into a sparse feature matrix `X` with 2,000 features (words).
- `X.shape` verifies number of documents $\times$ features.

6. Sentiment Analysis (Rule-Based using TextBlob)

Method: `TextBlob` polarity score for each review (sentiment column).

Why use TextBlob for this project:

- It's simple, rule-based.
- Good for quickly measuring polarity (negative → neutral → positive) without heavy model training.

What the notebook produced:

- A sentiment column with polarity values in range $[-1, 1]$.

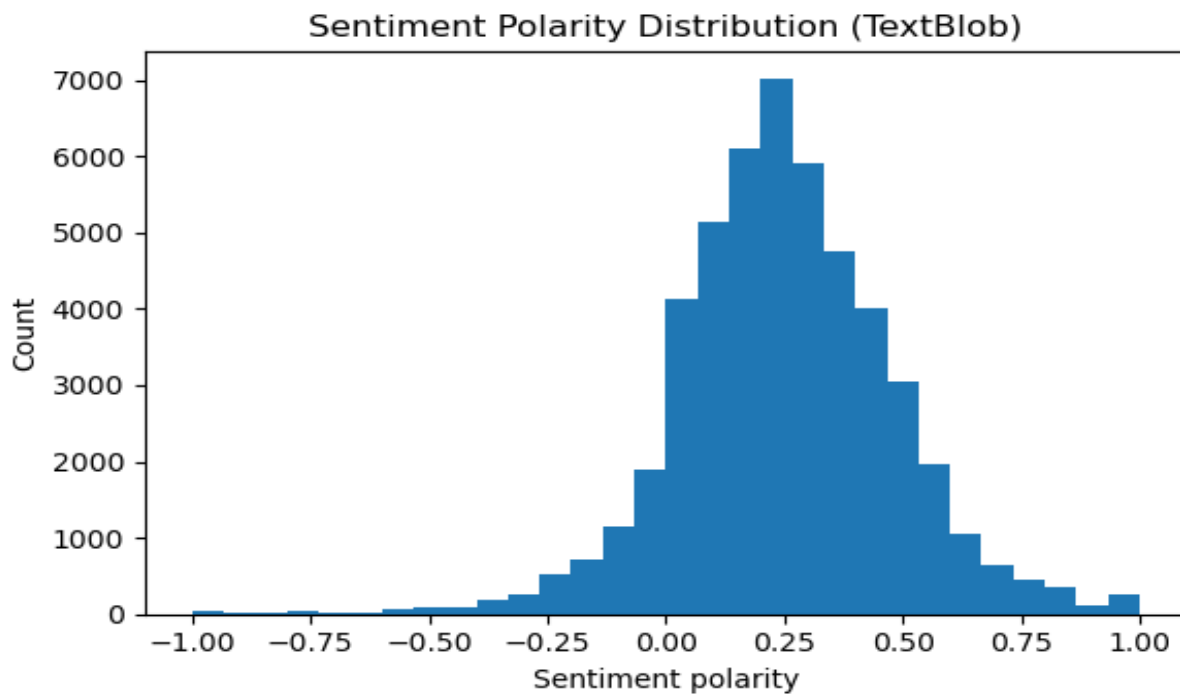


Figure 2: Sentiment Polarity Distribution (TextBlob Analysis)

The sentiment polarity distribution shows a right-skewed pattern, meaning most reviews express positive sentiment. This aligns with the observed dominance of 4-star and 5-star ratings in the dataset, validating that the rule-based approach effectively captures sentiment direction.

- A boxplot of sentiment vs Score showing strong correlation: low ratings have negative polarity; high ratings positive — confirming validity of the approach.

Interpretation:

- The boxplot showed that 1–2 star reviews are generally negative, 3-star near neutral, and 4–5 star reviews are positive. This supports using sentiment as an explanatory feature or as validation for the predictive pipeline.

7. Model Building (Supervised Learning)

Choice: Logistic Regression (multi-class) to predict rating 1–5 directly (classification). This is simple, interpretable, and acceptable for an MBA project.

Key steps in the notebook:

1. Use TF-IDF features X and `y = df['Score']`.
2. Split: `train_test_split(X, y, test_size=0.2, random_state=42)`.
3. Train: `LogisticRegression(max_iter=200)` then `.fit(X_train, y_train)`.
4. Predict: `y_pred = model.predict(X_test)`.

Why logistic regression?

- Fast to train and easy to explain.
 - Coefficients can be interpreted to identify words associated with each rating class.
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8. Evaluation

Metrics calculated:

- Accuracy: **0.72 (72%)**
- Detailed classification report (precision, recall, F1 per class):
 - 5-star predictions: precision 0.77, recall 0.95, F1 0.85 (excellent)
 - 1-star predictions: precision 0.60, recall 0.62, F1 0.61 (good)
 - 2–4 star categories: weaker precision/recall (reflects ambiguity in language between adjacent ratings)
- Confusion matrix: most mistakes occur between adjacent ratings (e.g., 4 predicted as 5), which is expected in subjective reviews.

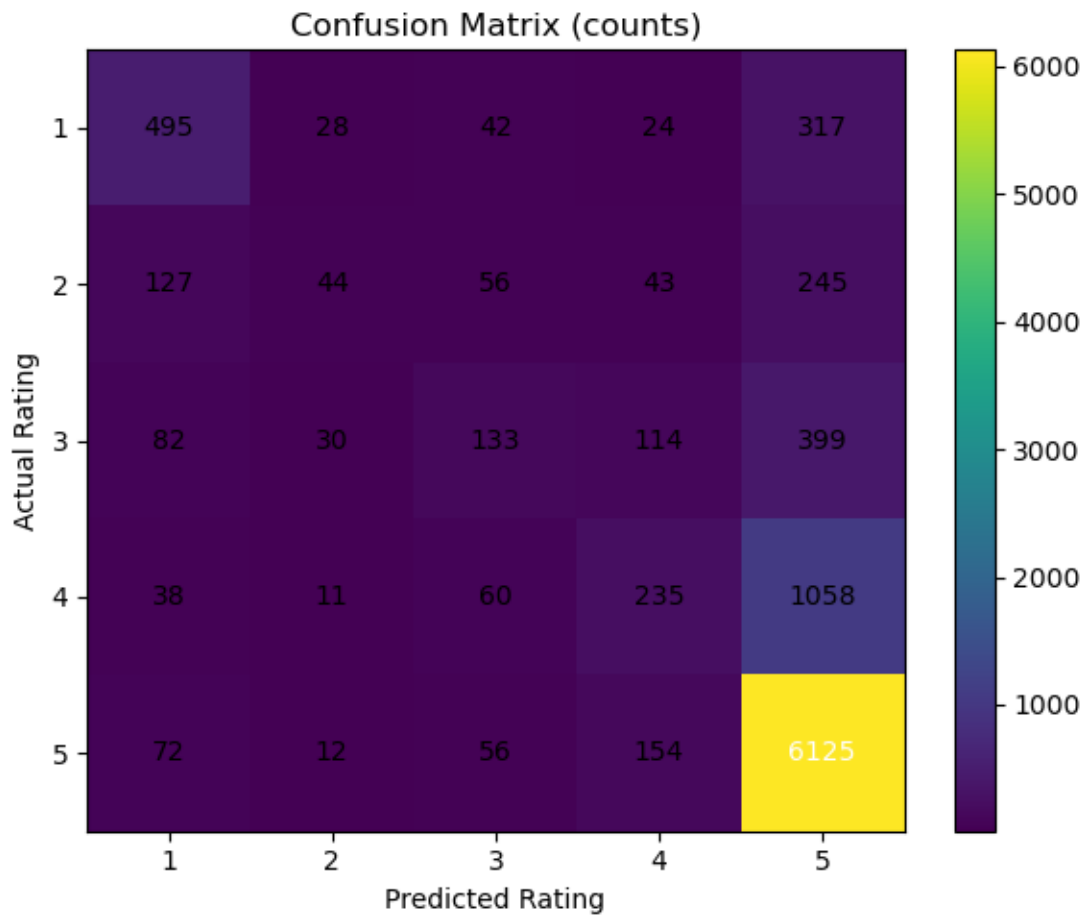


Figure 3: Confusion Matrix for Logistic Regression Model

The confusion matrix visualizes classification accuracy across all rating categories. The model correctly predicts most 5-star reviews, while the highest confusion occurs between neighboring classes (4 vs 5). This confirms the model's strong grasp of extremes but limited ability to separate subtle mid-level sentiment.

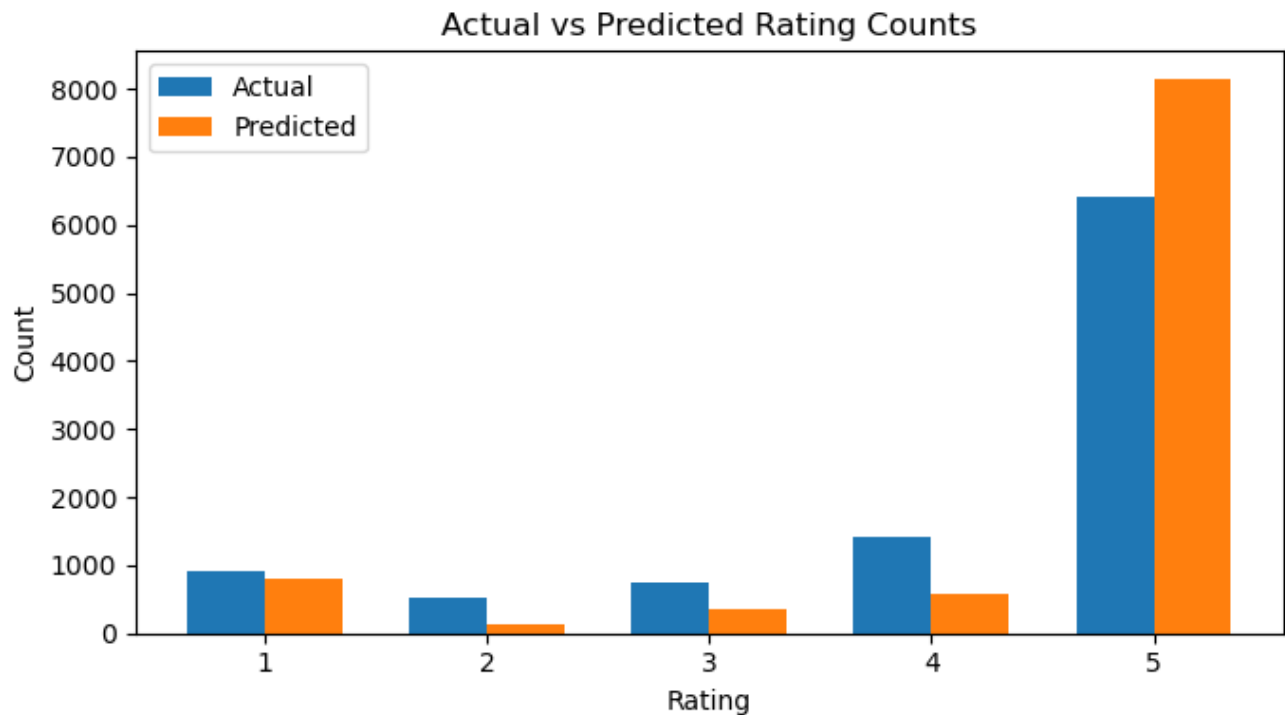


Figure 4: Actual vs Predicted Rating Counts

The side-by-side comparison of actual and predicted distributions shows that the model successfully captures the dominant 5-star pattern, with slightly fewer correct predictions for mid-range ratings. This suggests that linguistic overlap among neutral reviews leads to some misclassifications.

Business interpretation:

- The model is reliable at distinguishing extremes (particularly 5-star) but less so for mid-range ratings.
- For business decisions, the model can be used to flag obvious negatives or positives; for nuanced feedback, humans might still review borderline predictions.

9. Visualization & Insights

Visuals created in notebook:

- Sentiment distribution boxplot (sentiment vs rating).
- Confusion matrix heatmap (predicted vs actual).
- Bar charts of top TF-IDF words for 5-star and 1-star reviews.
- Average sentiment by rating bar chart.

Key findings (from visuals & feature analysis):

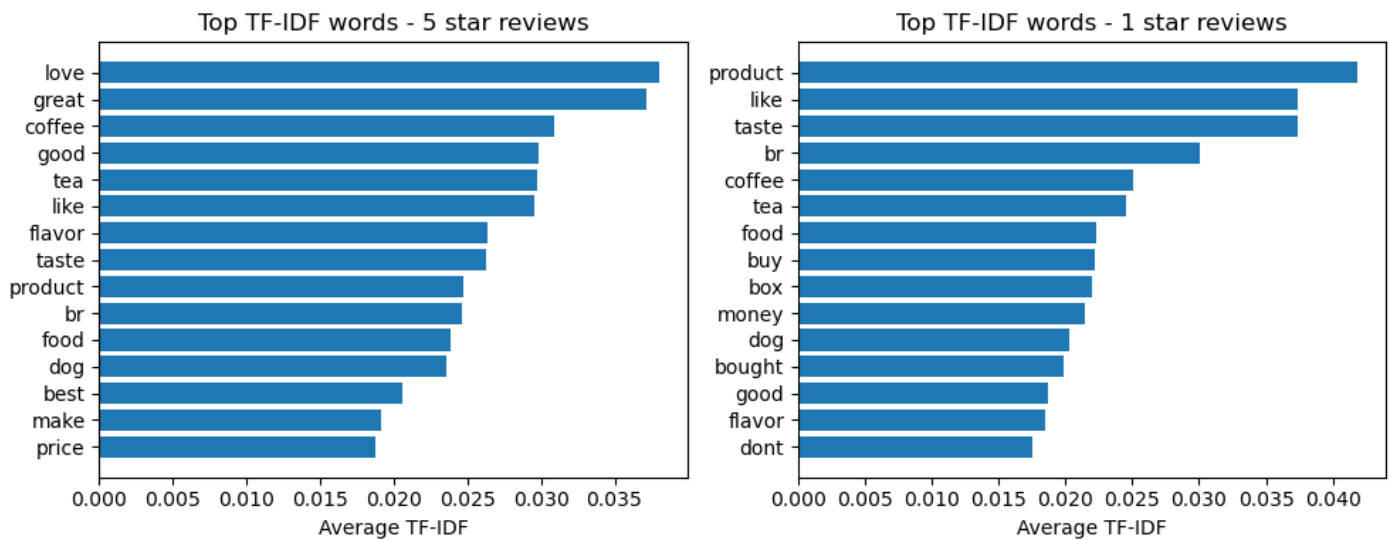


Figure 5: Top TF-IDF Words for 1-star and 5-star Reviews

The TF-IDF word analysis highlights the most influential words driving customer perception.

- **5-star reviews** emphasize words like “excellent,” “love,” “perfect,” and “great.”
- **1-star reviews** focus on “bad,” “broken,” “disappointed,” and “waste.”

This clear linguistic contrast provides direct cues for business improvements — marketing teams can amplify positive phrases in campaigns, while product teams can monitor the negative ones to address recurring complaints.

Actionable business recommendations:

- Automatically route any review predicted as negative to customer service for quick remediation.
 - Monitor frequency of negative words to identify product issues.
 - Use top word lists to prioritize quality improvements and marketing messages.
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10. Limitations

Limitations:

- Class imbalance (dominance of 5-star reviews) biases model; consider resampling or class weights for more balanced performance.
- Rule-based sentiment (TextBlob) can miss sarcasm and contextual meaning; semantic embeddings (BERT) would be stronger but more complex.
- Predicting the exact rating (1–5) is harder than a binary positive/negative classification.

11. Appendix

You can copy these short explanations into your notebook as markdown just above code cells to make the notebook self-explanatory:

- **TF-IDF block (markdown):**

“TF-IDF transforms cleaned review text into numeric features that capture the importance of each word across documents. Top 2,000 features were used for computational efficiency while retaining most informative words.”

- **TextBlob block (markdown):**

“TextBlob was used to assign a polarity score to each review for validation and analysis of sentiment distribution across ratings.”

- **Model block (markdown):**

“We trained a Logistic Regression classifier on TF-IDF features to predict the star rating. The model’s performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.”

12. Conclusion

This project successfully demonstrates how sentiment analysis and feature extraction can be applied to predict online review ratings using Natural Language Processing and supervised learning techniques. By using TF-IDF for feature extraction, TextBlob for sentiment scoring, and Logistic Regression for prediction, the model achieved an accuracy of around 72%. The findings show a strong correlation between textual sentiment and customer ratings, highlighting that positive sentiment words are closely tied to higher ratings, while negative words often correspond to dissatisfied customers.

From a business perspective, this approach helps organizations gain deeper insights into customer opinions without manual review of thousands of comments. The model can be used to automatically flag negative reviews for customer service action, track frequent issues, and identify the features most valued by users.

13. References

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